Signale & Systeme

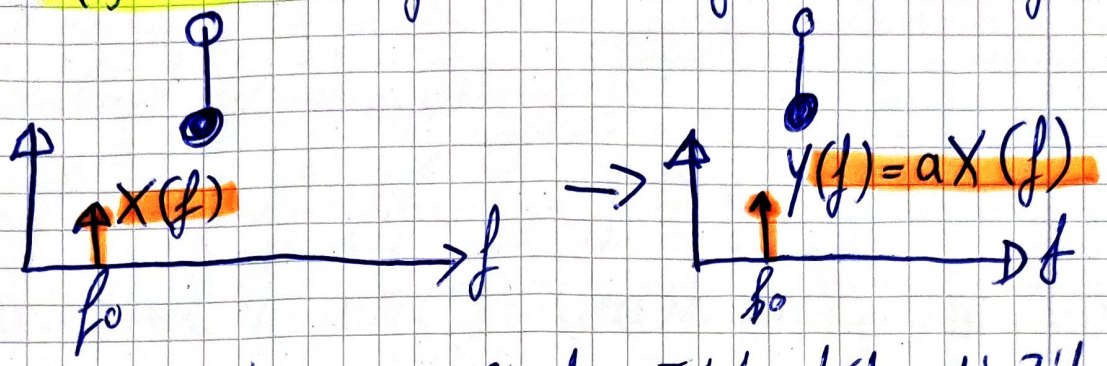
Linear-Time-Invariant (LTI)-Systems

Linearity: $x_i(t) \Rightarrow y_i(t)$

$$\sum_{i=0}^{N} a_i x_i(t) \Rightarrow \sum_{i=0}^{N} a_i y_i(t)$$

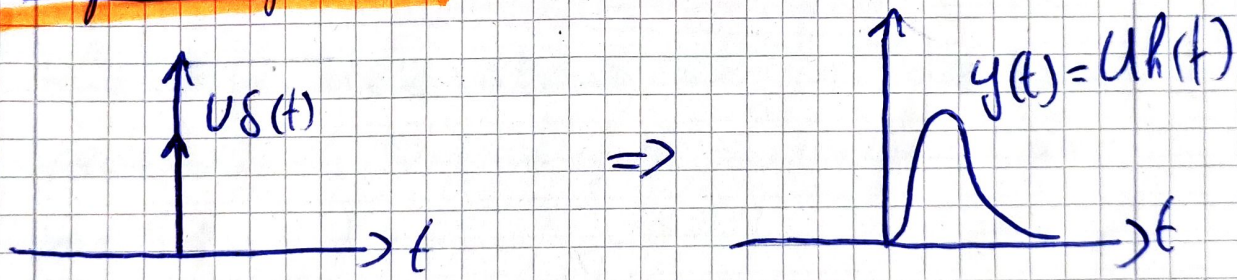
Linear amplifier:

$x(t) = \cos(2\pi f_0 t) \rightarrow y(t) = a\cos(2\pi f_0 t)$



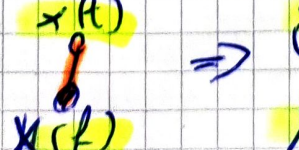
Time-Invariance: Signal verändert sich über die Zeit hinweg nicht. Does not depend on shift.

Impulse Response:



The impulse response $h(t)$ of an LTI system is given by the (normalized) output signal $y(t)$ / U if the input $x(t) = U\,\delta(t)$ is a (normalized) Dirac impulse

An LTI system is characterized by its impulse response $h(t)$ in the time domain or equivalent by its frequency response $H(f)$ in the frequency domain.



$$\Rightarrow \quad y(t) = h(t) * x(t)$$

$$Y(f) = H(f)\,X(f)$$

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Convolution

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$(-(\tau - t))$

Convolution with step function

$$y(t) = x(t) * \varepsilon(t) = \int_{-\infty}^{\infty} x(\tau)\, \underbrace{\varepsilon(t-\tau)}_{=0 \; t<\tau} d\tau = \int_{-\infty}^{t} x(\tau)\, d\tau$$

Properties of the Convolution:

A) $x(t) * y(t) = y(t) * x(t)$

B) $x(t) * [y(t) + z(t)] = x(t) * y(t) + x(t) * z(t)$

C) $x(t) * [y(t) * z(t)] = [x(t) * y(t)] * z(t)$

D) $x(t) * \delta(t) = x(t)$ Neutrales Element

Cyclic Convolution for Periodic Signals

For two periodic signals $x(t)$ and $y(t)$ with the same period T, the integrand of the convolution is also periodic

$$x(t) \circledast y(t) = \int_{t_0}^{t_0+T} x(\tau)\, y(t-\tau)\, d\tau$$

Shifted Dirac

$$x(t) * \delta(t-t_0) = x(t-t_0)$$

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Convolution with Dirac Impulse Comb

A periodic signal $x_T(t)$ can be represented as convolution of one period $x(t)$ of the signal with a Dirac impulse sequence:

$$x_T(t) = x(t) * \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

Correlation Function & Convolution

$$x \star (t) * y(t) = \int_{-\infty}^{\infty} x(\tau)\, y(t-\tau)\, d\tau \quad \text{or} \quad x(k) * y(k) = \sum_{i=-\infty}^{\infty} x(i)\, y(k-i)$$

$$r_{xy}^{E}(\tau) = x^*(-\tau) * y(\tau) \quad \text{or} \quad r_{xy}^{E}(k) = x^*(-k) * y(k)$$

⇒ A convolution has to be executed with the reflected signal!

$$r_{xy}^{P}(\tau) = \frac{1}{T}\, x(\tau) \circledast y(\tau)$$

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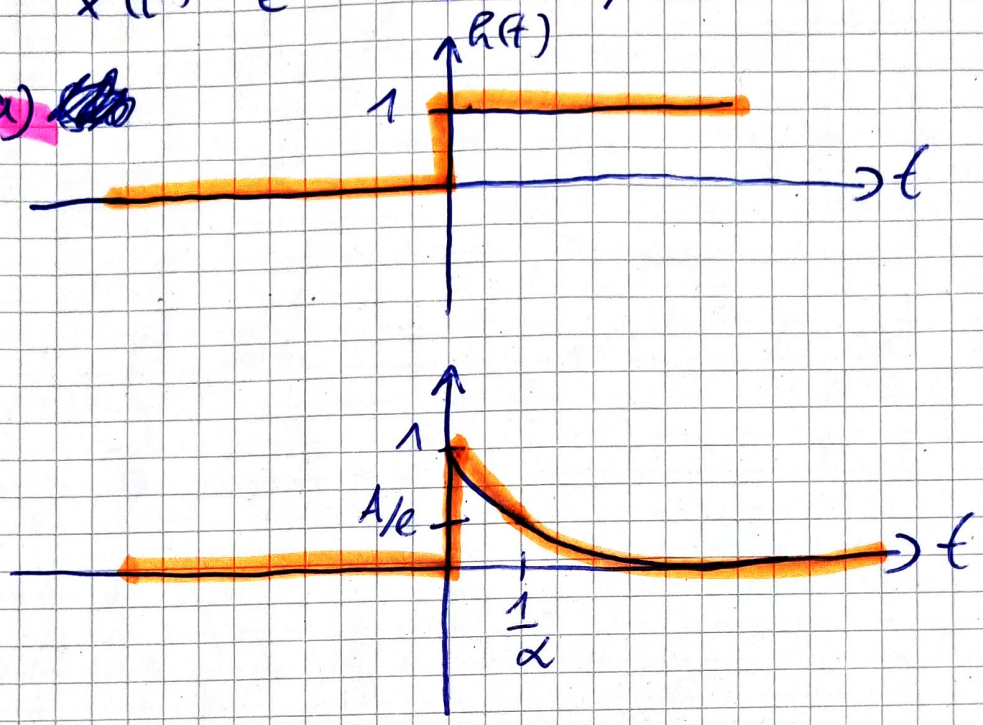
Convolution (1)

An LTI system is considered which has the impulse response $h(t)$. Furthermore, the input signal $x(t)$ is given. The signals are defined as follows

$$h(t) = \varepsilon(t)$$

$$x(t) = e^{-\alpha t} \cdot \varepsilon(t) \quad , \quad \alpha > 0$$

a)



b) Is the LTI system causal.

Yes because it depends only on $t \geq 0$

because

$$h(t) = 0 \quad , \quad t < 0$$

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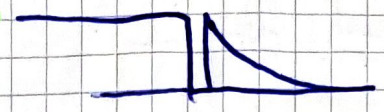
Convolution (1)

$h(-(\tau - t))$

c) $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t - \tau)\, d\tau$

$t < 0$:



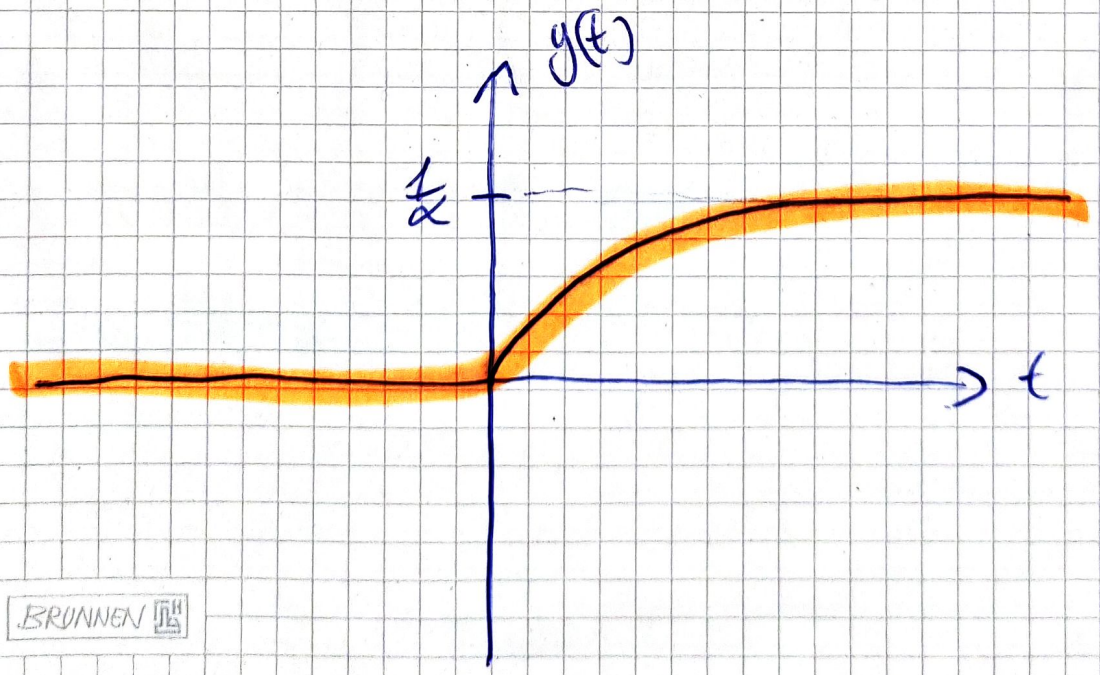
no overlap

$y(t < 0) = 0$

$t \geq 0$:

$$y(t) = \int_0^t e^{-\alpha t} = \left[\frac{1}{-\alpha} e^{-\alpha t} \right]_0^t = \frac{1}{\alpha} - \frac{1}{\alpha} e^{-\alpha t}$$

$$y(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{\alpha} - \frac{1}{\alpha} e^{-\alpha t}, & t \geq 0 \end{cases}$$



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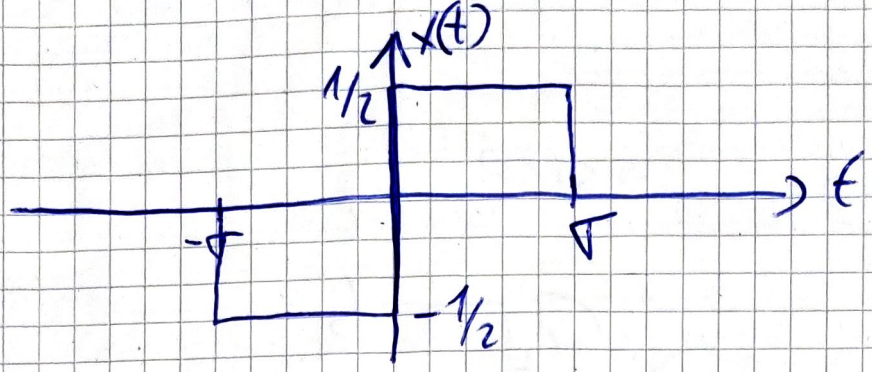
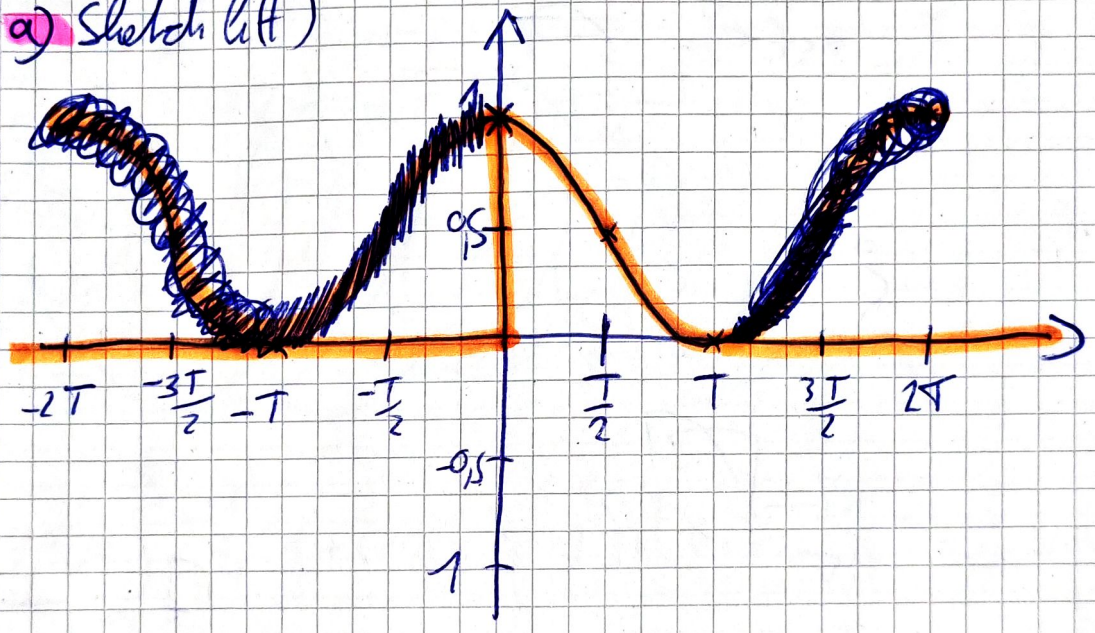
Convolution (3)

$$h(t) = \begin{cases} \frac{1}{2}\left(1+\cos\left(\frac{\pi t}{T}\right)\right), & 0 \le t \le T \\ 0, & \text{else} \end{cases}$$



a) Sketch h(t)



b) Is the LTI system causal?

Yes, because $h(t<0) = 0$

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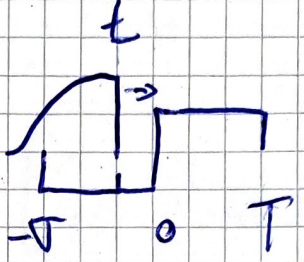
Faltung (3)

c) $y(t) = x(t) * h(t)$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$t \leq -T$: $y(t) = 0$



$-T \leq t \leq 0$:

$$y(t) = \int_{-T}^{t} -\frac{1}{4}\left(1 + \cos\left(\frac{\pi\tau}{T}\right)\right) d\tau = -\frac{1}{4}\tau - \frac{T}{4\pi}\sin\left(\frac{\pi\tau}{T}\right)\Big|_{-T}^{t}$$

$$= -\frac{1}{4}t - \frac{T}{4\pi}\sin\left(\frac{\pi t}{T}\right) + \frac{T}{4} - \cancel{\frac{T}{4\pi}\sin\left(\frac{\pi T}{T}\right)}^{0}$$

$$= -\frac{1}{4}\left(t + T + \frac{T}{\pi}\sin\left(\frac{\pi t}{T}\right)\right)$$

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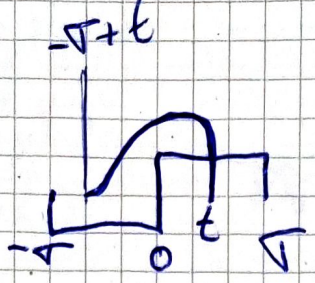
Faltung (3) c)

c)

$0 \leq t < T$:



$$y(t) = \int_{-T+t}^{0} (-1)\left(\frac{1}{4}\left(1+\cos\left(\frac{\pi\tau}{T}\right)\right)\right) d\tau + \int_{0}^{t} \frac{1}{4}\left(1+\cos\left(\frac{\pi\tau}{T}\right)\right) d\tau$$

$$= \left[-\frac{1}{4}\tau - \frac{T}{4\pi}\sin\left(\frac{\pi\tau}{T}\right)\right]_{-T+t}^{0} + \left[\frac{1}{4}\tau + \frac{T}{4\pi}\sin\left(\frac{\pi\tau}{T}\right)\right]_{0}^{t}$$

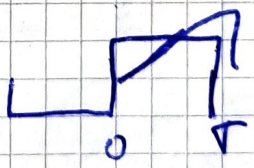
$$= +\frac{1}{4}\left(-T+t\right) + \frac{T}{\pi}\sin\left(\frac{-T\pi+t\pi}{T}\right) + \frac{1}{4}t + \frac{T}{4\pi}\sin\left(\frac{\pi t}{T}\right)$$

$\sin(\alpha-180°) = -\sin(\alpha)$

$$= -\frac{1}{4}T + \frac{1}{4}t + \frac{1}{4}t + \frac{T}{4\pi}\sin\left(\frac{\pi t}{T} - \pi\right) + \frac{T}{4\pi}\sin\left(\frac{\pi t}{T}\right)$$

$$= -\frac{1}{4}T + \frac{1}{2}t + \frac{T}{4\pi}\sin\left(\frac{\pi t}{T}\right) + \frac{T}{4\pi}\sin\left(\frac{\pi t}{T}\right)$$

$T \leq t \leq 2T$:



$$y(t) = \int_{-T+t}^{T} \frac{1}{4}\left(1+\cos\left(\frac{\pi\tau}{T}\right)\right) d\tau$$

$$= \left[\frac{1}{4}\tau + \frac{T}{4\pi}\sin\left(\frac{\pi\tau}{T}\right)\right]_{-T+t}^{T} = \frac{1}{4}T + 0 - \frac{1}{4}(-T+t) - \frac{T}{4\pi}\sin\left(\frac{\pi(-T+t)}{T}\right)$$

$$= \frac{T}{2} - \frac{t}{4} - \frac{T}{4\pi}\sin\left(\frac{\pi\left(-T+t\right)}{T}\right)$$

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$t > 2T$: $y(t) = 0$

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Cyclic Convolution

$$x(t) = \cos(2\pi f_0 t)$$
$$y(t) = \sin(2\pi f_0 t) \qquad T = \frac{2\pi}{2\pi f_0} = \frac{1}{f_0}$$

a)

$$x(t) \circledast y(t) =$$

$$\sin(\alpha)\cos(\beta) = \frac{1}{2}(\sin(\alpha-\beta) + \sin(\alpha+\beta))$$

$$\int_0^{1/f_0} x(\tau)\, y(t-\tau)\, d\tau$$

$$= \int_0^{1/f_0} \cos(2\pi f_0 \tau) \sin(2\pi f_0 (t-\tau))\, d\tau$$

$$= \int_0^{1/f_0} \frac{1}{2}\left(\sin(2\pi f_0(t-\tau-\tau)) + \sin(2\pi f_0(t-\tau+\tau))\right) d\tau$$

$$= \int_0^{1/f_0} \frac{1}{2}\left(\sin(2\pi f_0(t-2\tau)) + \sin(2\pi f_0 t)\right) d\tau$$

integration over two periods

$$= \frac{1}{2}\sin(2\pi f_0 t) \cdot \tau \Big|_0^{1/f_0}$$

$$= \frac{1}{2}\sin(2\pi f_0 t)\frac{1}{f_0}$$

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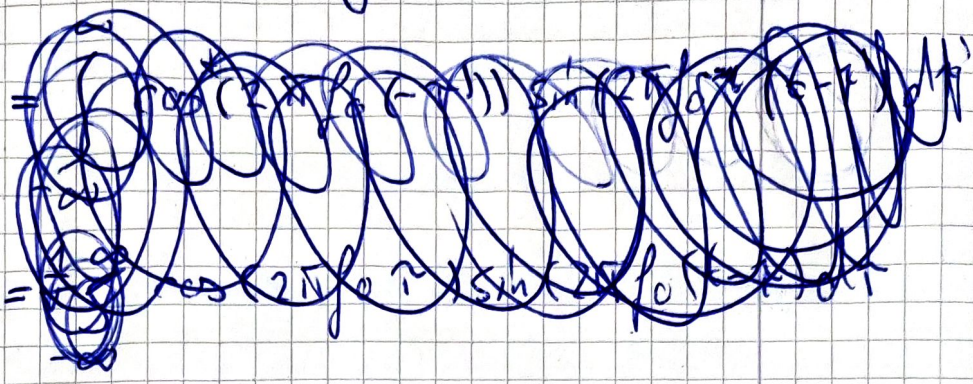
Cyclic Correlation

b) $r_{xy}^{P}(\tau) = \frac{1}{T} x^{*}(-T) \circledast y(T)$



$= \frac{1}{2} \sin(2\pi f_0 t)$

c)

$r_{xy}^{P}(0) = 0 \quad \rightsquigarrow \cos \perp \sin$

$r_{xy}^{P}\left(\frac{T}{4}\right) = \frac{1}{2} \sin\left(\frac{\pi f_0}{2 f_0}\right) = \frac{1}{2} \rightsquigarrow$ correlation of two identical signals yield the power

$r_{xy}^{P}\left(\frac{3T}{4}\right) = \frac{1}{2} \cdot \sin\left(\frac{3\pi}{2}\right) = -\frac{1}{2} \rightsquigarrow$

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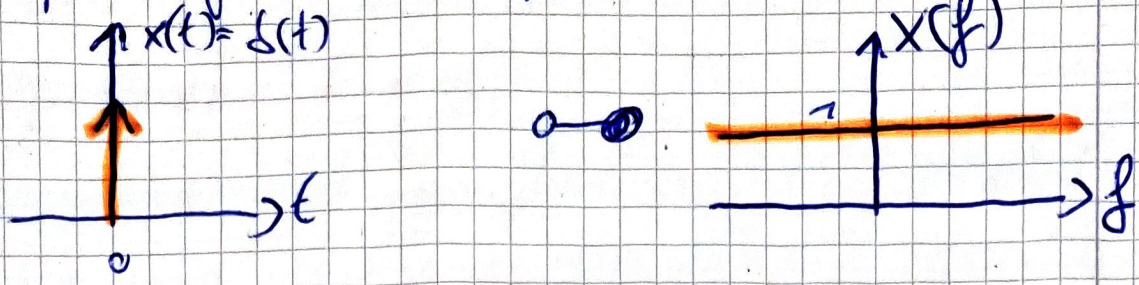
Properties of an LTI System



An LTI system can be characterized by its impulse response

The impulse response h(t) characterizes the output signal of a system when the input signal is a Dirac impuls

Causal Systems:

The output signal $y(t_0)$ of a causal system at time t_0 does not depend on future input values $x(t > t_0)$

↳ $h(t) = 0 \; ; \; t < 0$

↳ $y(t_0) = H\{x(t \leq t_0)\}$

Stable System:

Bounded Input Bounded Output (BIBO) Stability:

A system is called stable if, the response to a bounded input signal; i.e. $|x(t)| < \infty$ is a bounded output signal $|y(t)| < \infty$

$$\Rightarrow \int_{-\infty}^{\infty} |h(\tau)| \, d\tau < \infty$$

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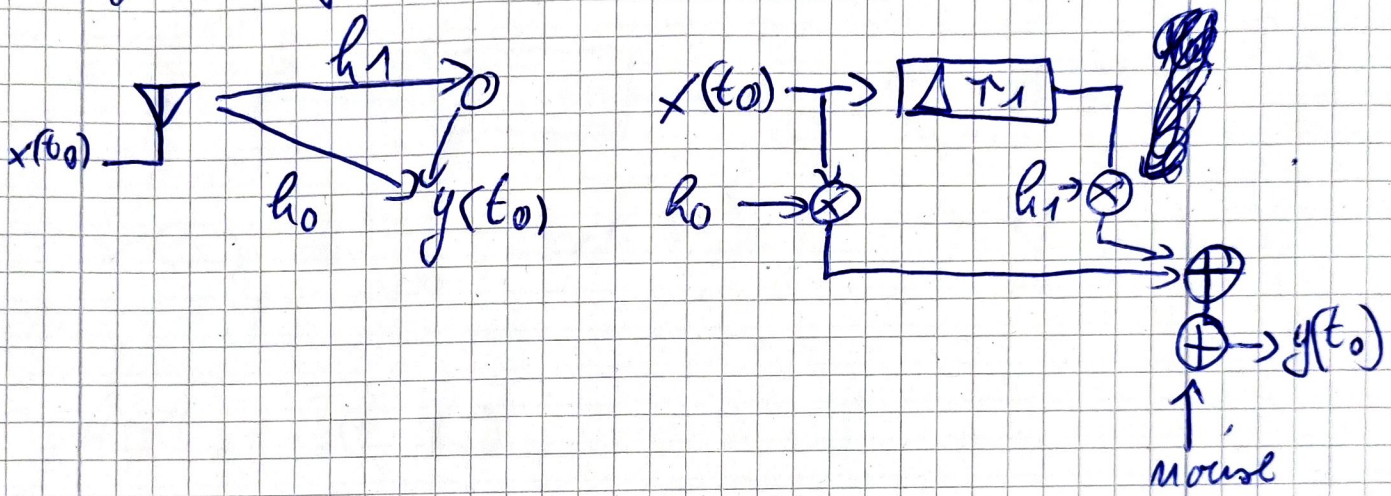
Properties of ΔT_1 - System

Memoryless

A system is called memoryless or static if the output signal $y(t_0)$ at time t_0 depends only on the value $x(t_0)$ of the input signal at the same time t_0.

Otherwise it is called dynamic system.

dynamic eg:



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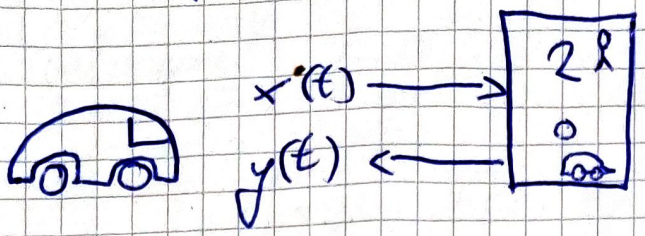
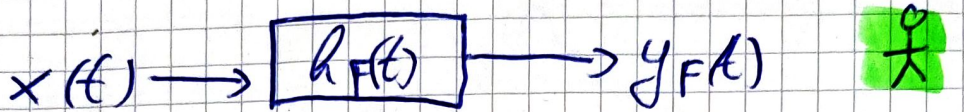
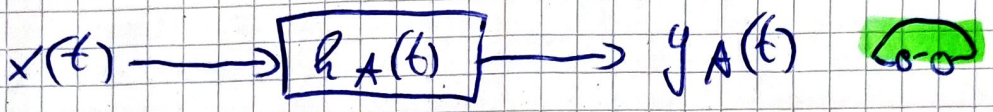
Pedestrian Recognition



pedestrian: $x(t) * h_F(t) \rightarrow y_F(t)$

car: $x(t) * h_A(t) \rightarrow y_A(t)$





$$x(t) = \frac{1}{\sqrt{\tau}} \text{rect}\left(\frac{t}{\tau}\right) \quad ; \tau > 0$$

$$h_F(t) = \delta(t - t_0)$$

$$h_A(t) = \frac{1}{\sqrt{3}} \left[\delta(t - t_0 + \tau) + \delta(t - t_0) + \delta(t - t_0 - \tau)\right]$$

$$t_0 >> \tau$$

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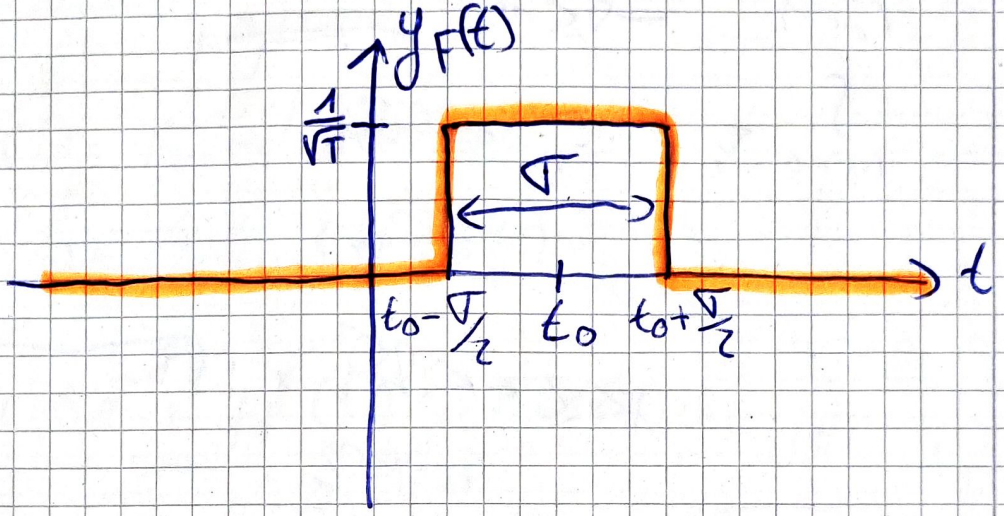
Pedestrian Recognition

a) Determine the receive signal $y_F(t)$

$$y_F(t) = h_F(t) * x(t) = \delta(t - t_0) * x(t)$$

$$= x(t - t_0) = \frac{1}{\sqrt{\nabla}} \operatorname{rect}\left(\frac{t - t_0}{\nabla}\right)$$

b)



c) Determin the receive signal $y_A(t)$

$$y_A(t) = h_A(t) * x(t) = \frac{1}{\sqrt{3}} \left(\delta(t - t_0 + \nabla) * x(t) + \delta(t - t_0) * x(t) + \delta(t - t_0 - \nabla) * x(t) \right)$$

$$= \frac{1}{\sqrt{3}} x(t - t_0 + \nabla) + \frac{1}{\sqrt{3}} x(t - t_0) + \frac{1}{\sqrt{3}} x(t - t_0 - \nabla)$$

$$= \frac{1}{\sqrt{3\nabla}} \operatorname{rect}\left(\frac{t - t_0 + \nabla}{\nabla}\right) + \frac{1}{\sqrt{3\nabla}} \operatorname{rect}\left(\frac{t - t_0}{\nabla}\right) + \frac{1}{\sqrt{3\nabla}} \operatorname{rect}\left(\frac{t - t_0 - \nabla}{\nabla}\right)$$

$$= \frac{1}{\sqrt{3\nabla}} \operatorname{rect}\left(\frac{t - t_0}{3\nabla}\right)$$

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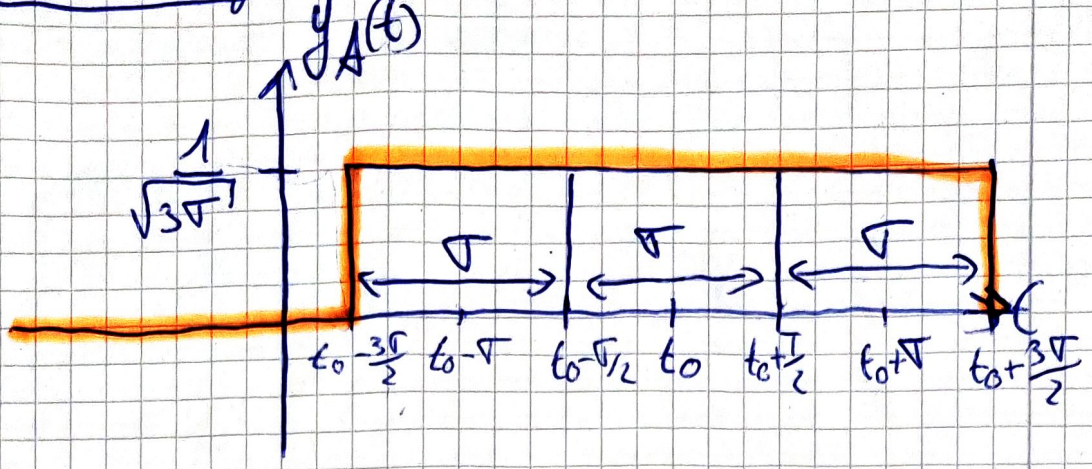
d)

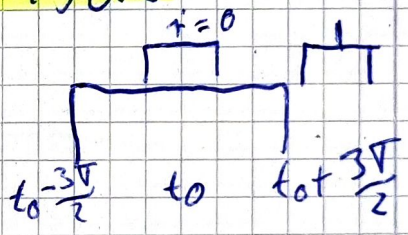


e) Determine $r_{y_A y_F}(\tau)$ with

$$y_F(t) = \frac{1}{\sqrt{T}} \operatorname{rect}\left(\frac{t-t_0}{T}\right)$$

$$y_A(t) = \frac{1}{\sqrt{3T}} \operatorname{rect}\left(\frac{t-t_0}{3T}\right)$$

$$r_{y_A y_F}(\tau) = \int_{-\infty}^{\infty} y_A^*(t)\, y_F(t+\tau)\, dt$$



$\tau < -2T$: $r_{y_A y_F}(\tau) = 0$

$\tau > 2T$: $r_{y_A y_F}(\tau) = 0$

Achte darauf bei correlation } Desweg in Grau - τ

$\infty \longleftarrow -\infty$

α bei Faltung

$-\infty \longrightarrow +\infty$

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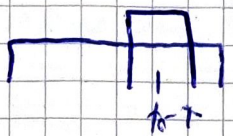
e)

$-2\nabla \leq \tau < -\nabla$:

$$r_{y_A y_F} = \int_{t_0-\tau-\nabla/2}^{t_0+\frac{3\nabla}{2}} \frac{1}{\sqrt{3}\nabla}\, dt = \frac{t_0+\frac{3\nabla}{2} - t_0+\tau+\nabla/2}{\sqrt{3}\nabla}$$

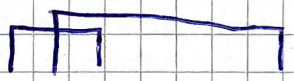
$$= \frac{\tau+2\nabla}{\sqrt{3}\nabla}$$



$-\nabla \leq \tau \leq \nabla$:

$$r_{y_A y_F} = \int_{t_0-\tau-\nabla/2}^{t_0-\tau+\nabla/2} \frac{1}{\sqrt{3}\nabla}\, dt = \frac{t_0-\tau+\nabla/2 - t_0+\tau+\nabla/2}{\sqrt{3}\nabla}$$

$$= \frac{1}{\sqrt{3}}$$



$\nabla < \tau \leq 2\nabla$:

$$r_{y_A y_F} = \int_{t_0-\frac{3\nabla}{2}}^{t_0-\tau+\nabla/2} \frac{1}{\sqrt{3}\nabla}\, dt = \frac{t_0-\tau+\nabla/2 - t_0+\frac{3\nabla}{2}}{\sqrt{3}\nabla}$$

$$= \frac{2\nabla-\tau}{\sqrt{3}\nabla}$$

$$r_{y_A y_F}(\tau) = \begin{cases} 0 & , |\tau| > 2\nabla \\ \frac{1}{\sqrt{3}\nabla}(\tau+2\nabla) & , -2\nabla \leq \tau < -\nabla \\ 1/\sqrt{3} & , -\nabla \leq \tau < \nabla \\ \frac{1}{\sqrt{3}\nabla}(-\tau+2\nabla) & , \nabla \leq \tau \leq 2\nabla \end{cases}$$

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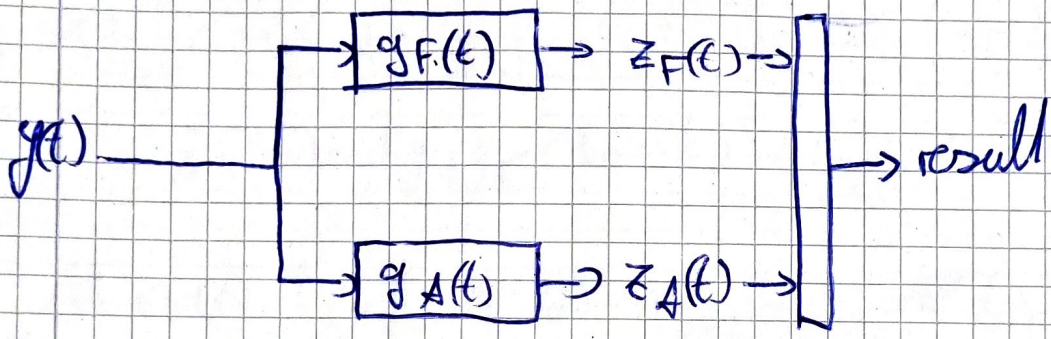
Pedestrian Recognition

$r_{y_F y_F}(\tau) = \Lambda\left(\frac{\tau}{2T}\right)$

$r_{y_A y_A}(\tau) = \Lambda\left(\frac{\tau}{3T}\right)$

a) $r_{y_F y_F}(0) = 1$; same signal, $E_{y_F} = r_{y_F y_F}(0)$

$r_{y_A y_A}(0) = 1$; same signal, $E_{y_A} = r_{y_A y_A}(0)$



$g_F(t) = y_F^*(t_0 - t)$; $g_A(t) = y_A^*(t_0 - t)$

$z_F(t) = r_{y_F y_F}(t - t_0)$

$z_A(t) = r_{y_A y_F}(t - t_0)$

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Pedestrian Recognition:

g) prove that $z_A(t) = r_{y_A y_F}(t - t_0)$

$$z_A(t) = y_F(t) * y_A^*(t_0 - t) = \int_{-\infty}^{\infty} y_F(\tau)\, y_A^*(\underbrace{t_0 - (t-\tau)}_{=\tau'})\, d\tau$$

$$= \int_{-\infty}^{\infty} y_F(\tau' + t - t_0)\, y_A^*(\tau')\, d\tau'$$

$$= r_{y_A y_F}(t - t_0) \qquad \square$$

h) $z_F(t_0) = r_{y_F y_F}(t_0 - t_0) = \Delta\left(\frac{0}{T}\right) = 1$

$$z_A(t_0) = r_{y_A y_F}(t_0 - t_0) = r_{y_A y_F}(0) = \frac{1}{\sqrt{3}}$$

i) $z_F(t) = r_{y_F y_A}(t - t_0) = r_{y_A y_F}^*(t_0 - t)$

$$z_A(t) = r_{y_A y_A}(t - t_0)$$

$$z_F(t_0) = r_{y_A y_A}(t_0 - t_0) = \frac{1}{\sqrt{3}}$$

$$z_A(t_0) = 1$$

j) $\frac{1}{\sqrt{3}} \rightarrow$ Pedestrian $\qquad 1 \rightarrow$ car

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Signale & Systeme

Varianten der Fourier Transform

Fourier Integral: continuous frequency & non-periodic } continuous time
Fourier-Series: discrete frequency & non-periodic
Discrete-Time Fourier Transform (DTFT): continuous frequency & periodic } discrete time
Discrete Fourier Transform: discrete frequency & periodic

Fourier Series:

Periodic time domain function with period T can be represented as sum of cos & sin oscillations! $f_0 = \frac{1}{T}$

The Fourier spectrum of a periodic time domain function is discrete!

The frequency-range of the Fourier spectrum is called the bandwidth of the signal!

Real Fourier Series:

$f_0 = \frac{1}{T}$

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(2\pi n f_0 t) + \sum_{n=1}^{\infty} B_n \sin(2\pi n f_0 t)$$

Fourier Coefficients:

$$A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(2\pi n f_0 t)\, dt \; ; \; B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(2\pi f_0 n t)\, dt$$

Complex Fourier Series:

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{j 2\pi n f_0 t} \qquad X_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j 2\pi n f_0 t}\, dt$$

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Signale & Systeme

Fourier Series

A Dirac $\delta(f-f_0)$ represented a complex oscillation $e^{j2\pi f_0 t}$.

↳ a cosine is represented by two Diracs since

$$\cos(2\pi f_0 t) = \frac{1}{2}\left(e^{2\pi f_0 t} + e^{2\pi(-f_0)t}\right)$$

- A periodic time domain signal $x(t)$ always has a discrete spectrum $X(f)$
- A non-periodic time domain signal $x(t)$ always has a continuous spectrum $X(f)$

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Signale & Systeme

Fourier Series

The complex Fourier series of a periodic function $x(t)$ is given by

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{j 2\pi n f_0 t} \quad \text{with complex function}$$

complex Fourier coefficients $X_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j 2\pi n f_0 t} dt$

5) Show that the complex Fourier coefficients X_n of a real periodic signal $x(t)$ are conjugate even

$$X_n = X_{-n}^*$$

$$X_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) (\cos(-2\pi n f_0 t) + j \sin(-2\pi n f_0 t)) dt$$

$$X_{-n} = \frac{1}{T} \int_{-T/2}^{T/2} x(t) (\cos(-2\pi n f_0 t) - j \sin(-2\pi n f_0 t)) dt$$

$$X_{-n}^* = \frac{1}{T} \int_{-T/2}^{T/2} x(t) (\cos(-2\pi n f_0 t) + j \sin(-2\pi n f_0 t)) dt$$

$$\hookrightarrow \quad X_n = X_{-n}^* \qquad \blacksquare$$

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Signale & Systeme

Fourier Series

c) Show that for real signals $x(t)$, an equivalent representation of the Fourier series is given by

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(2\pi n f_0 t) + \sum_{n=1}^{\infty} B_n \sin(2\pi n f_0 t)$$

$$A_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(2\pi n f_0 t)\,dt \quad \& \quad B_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(2\pi n f_0 t)\,dt$$

$$X_n = X_{n,r} + jX_{n,i}$$

useless proof!

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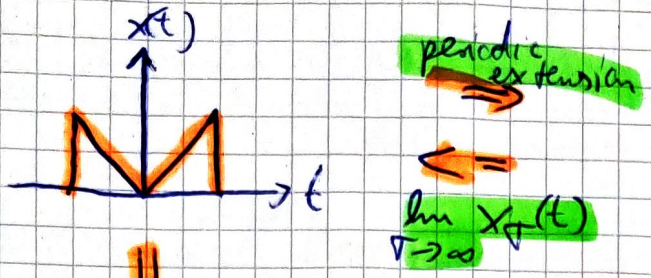
Signale & Systeme

Fourier Transform

Fourier Integral: non-periodic & continuous time

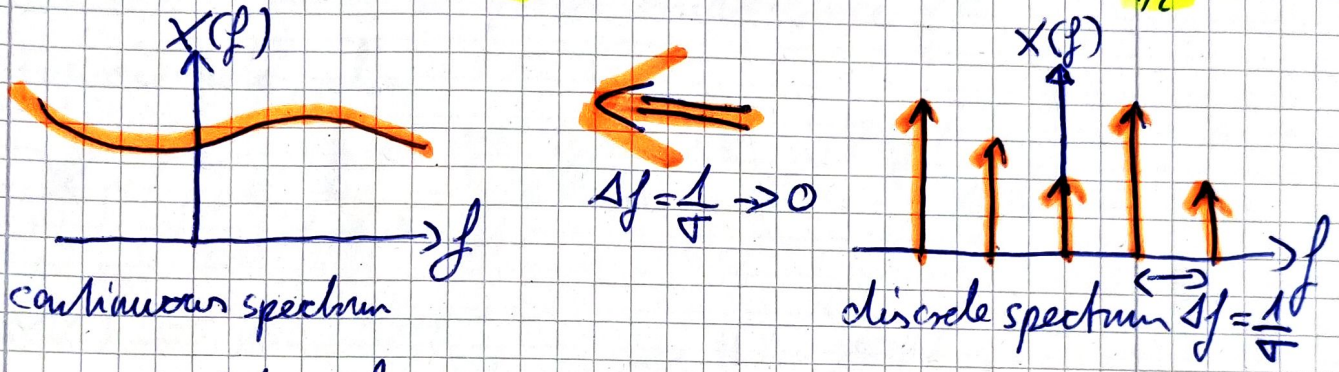


Fourier Integral

$$X(f) = \lim_{T\to\infty} T \cdot X_S(f) = \int_{-\infty}^{\infty} x(t) e^{-j\pi f t}\, dt$$

Fourier Series

$$X_n = X_S\left(f = \frac{n}{T}\right) = \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-j\pi n t/T}\, dt$$



Fourier Integral

The frequency domain represented of a non-periodic time domain signal is obtained by the Fourier Integral

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t}\, dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t}\, dt$$

!

$$x(t) \circ\!\!-\!\!\bullet\, X(f)$$

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Convergence of the Fourier Integral

Fourier integral exist if

$$\int_{-\infty}^{\infty} |x(t)|\,dt < \infty \Rightarrow \int_{-\infty}^{\infty} |x(t)|^2\,dt < \infty \rightarrow \text{energy limited signals}$$

↳ Problem with Power Signals !

Fourier Transform with Convergence Factor

The Fourier-Transform uses an attenuation Factor $e^{-\frac{1}{2}\alpha|t|}$ in order to force convergence of the Fourier integral.

$$X_F(f) = \lim_{\alpha \to 0} \int_{-\infty}^{\infty} x(t) e^{-\alpha|t|} e^{-j2\pi ft}\,dt$$

$$x(t) = \lim_{\alpha \to 0} \int_{-\infty}^{\infty} X_F(f) e^{-\alpha|t|} e^{j2\pi ft}\,dt$$

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Properties of Fourier-Transform

$$x(t) \circ\!\!-\!\!\bullet\; X(f)$$

Linearity:

$$\sum_i a_i x_i(t) \circ\!\!-\!\!\bullet\; \sum_i a_i X_i(f)$$

[illegible]

Duality:

$$X(t) \circ\!\!-\!\!\bullet\; x(-f)$$

$$\delta(t) \circ\!\!-\!\!\bullet\; 1$$

$$\delta(t-t_0) \circ\!\!-\!\!\bullet\; e^{-j2\pi f t_0}$$

$$e^{j2\pi f_0 t} \circ\!\!-\!\!\bullet\; \delta(f-f_0)$$

$$\cos(2\pi f_0 t) \circ\!\!-\!\!\bullet\; \frac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$$

$$\sin(2\pi f_0 t) \circ\!\!-\!\!\bullet\; \frac{j}{2}[\delta(f+f_0)-\delta(f-f_0)]$$

Convolution:

$$x(t) * y(t) \circ\!\!-\!\!\bullet\; X(f)\,Y(f)$$

$$x(t) \cdot y(t) \circ\!\!-\!\!\bullet\; X(f) * Y(f)$$

$$r_{xy}^{E}(\tau) \circ\!\!-\!\!\bullet\; X^*(f) \cdot Y(f)$$

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Properties of Fourier Transform

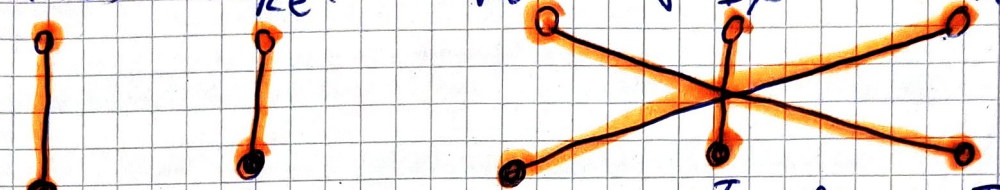
$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) \quad \circ\!\!-\!\!\bullet \quad X(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T}\right)$$

$$\varepsilon(t) \quad \circ\!\!-\!\!\bullet \quad \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$$

$$\mathrm{sgn}(t) \quad \circ\!\!-\!\!\bullet \quad \frac{1}{j2\pi f}$$

Symmetrie Properties:

$$x(t) = x_{Re}(t) + x_{Ro}(t) + j x_{Ie}(t) + j x_{Io}(t)$$



$$X(f) = X_{Re}(f) + X_{Ro}(f) + jX_{Ie}(f) + jX_{Io}(f)$$

$$x(-t) \quad \circ\!\!-\!\!\bullet \quad X(-f)$$

$$x^*(t) \quad \circ\!\!-\!\!\bullet \quad X^*(-f)$$

$$x^*(-t) \quad \circ\!\!-\!\!\bullet \quad X^*(f)$$

$$x(at) \quad \circ\!\!-\!\!\bullet \quad \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

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Fourier Transform - Time-Bandwidth Product

The time-bandwidth product $T \cdot B$ of an arbitrarily chosen interval T in time and an arbitrarily chosen interval B in frequency is not affected by temporal scaling.

$$x(at) \quad \circ\!\!-\!\!\bullet \quad \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

If a time domain signal is expanded by a factor a, the Fourier spectrum is compressed by the same factor!

Time - Limited Signals & Band - Limited Signals

A signal is called band-limited, if its spectrum is equal to zero for frequencies larger than a cut-off frequency f_c

if $X(f) = 0 \;, \; |f| \geq f_0$

↳ A time limited signal is never band limited

↳ A band limited signal is never time limited

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Bandwidth

1) Bandwidth B

Frequency range of spectrum at positive frequencies

Fourier Transform

$$x(t-t_0) \quad \circ\!\!-\!\!\bullet \quad X(f)\, e^{-j2\pi f t_0}$$

$$e^{j2\pi f_0 t}\, x(t) \quad \circ\!\!-\!\!\bullet \quad X(f-f_0)$$

$$\frac{d}{dt}\, x(t) \quad \circ\!\!-\!\!\bullet \quad j2\pi f\, X(f)$$

$$-j2\pi t\, x(t) \quad \circ\!\!-\!\!\bullet \quad \frac{d}{df}\, X(f)$$

$$\int_{-\infty}^{t} x(\tau)\, d\tau \quad \circ\!\!-\!\!\bullet \quad \frac{1}{2} X(0)\, \delta(f) + \frac{1}{j2\pi f}\, X(f)$$

$$\int_{-\infty}^{\infty} x(t)\, y^*(t)\, dt \quad \circ\!\!-\!\!\bullet \quad \int_{-\infty}^{\infty} X(f) \cdot Y^*(f)\, df$$

$$\Downarrow$$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

Complex Fourier-Series:

$$x(t) = \sum_{n=-\infty}^{\infty} X_n\, e^{j2\pi n f_0 t}$$

$$\Downarrow$$

$$X(f) = \sum_{n=-\infty}^{\infty} X_n\, \delta(f - n f_0)$$

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Table of Fourier-Transform

$$\text{rect}\left(\frac{t}{\tau}\right) \circ\!\!-\!\!\bullet\; \tau\,\frac{\sin(\pi f \tau)}{\pi f \tau}$$

$$\text{si}\left(\frac{\pi t}{\tau}\right) \circ\!\!-\!\!\bullet\; X(f) = \begin{cases} \tau, & |f| < \frac{1}{2\tau} \\ \tau/2, & |f| = \frac{1}{2\tau} \\ 0, & \text{else} \end{cases}$$

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Fourier Transform

a) Show that holds

$$x(t) = \text{rect}\left(\frac{t}{T}\right) \circ\!\!-\!\!\bullet\, X(f) = |T|\,\text{si}(\pi f T)$$

$$\text{rect}(t) \circ\!\!-\!\!\bullet\, \text{si}(\pi f)$$

$$x(at) \circ\!\!-\!\!\bullet\, \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

$$\hookrightarrow \text{rect}\left(\frac{t}{T}\right) \circ\!\!-\!\!\bullet\, |T|\,\text{si}(\pi f T)$$

b) Derive the Fourier Transform of $x(t) = |F|\,\text{si}(\pi F t)$

$$|F|\,\text{si}(\pi F t) \circ\!\!-\!\!\bullet\, \frac{|F|}{|F|}\,\text{rect}(-f/F)$$

$$= \text{rect}\left(\frac{f}{F}\right) = \text{rect}(fT)$$

$$x(t) \circ\!\!-\!\!\bullet\, X(f)$$

$$X(t) \circ\!\!-\!\!\bullet\, x(-f)$$

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Fourier Transform

c) Calculate the inverse Fourier Transform

$$\underline{X}(f) = \underline{\Lambda}\left(\frac{f-f_0}{B}\right) \quad , B>0$$

$$si^2(\pi t) \circ\!\!-\!\!\bullet\; \Lambda(f)$$

$$e^{j2\pi f_0 t}\, x(t) \circ\!\!-\!\!\bullet\; X(f-f_0)$$

$$x(at) \circ\!\!-\!\!\bullet\; \frac{1}{|a|}\underline{X}\left(\frac{f}{a}\right)$$

$$|B|\, si^2(\pi B t)\, e^{j2\pi f_0 t} \circ\!\!-\!\!\bullet\; \underline{\Lambda}\left(\frac{f-f_0}{B}\right) \quad , B>0$$

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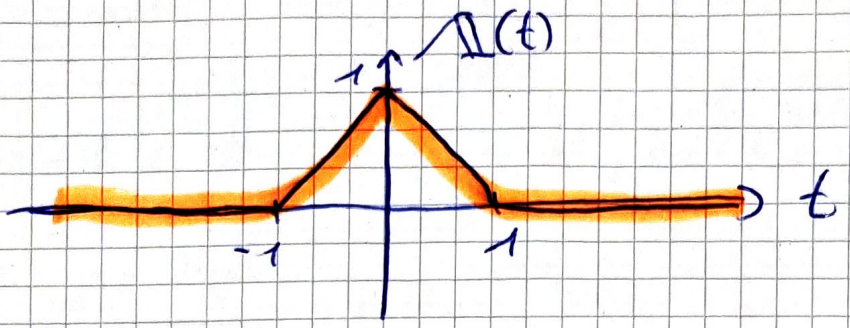
Fourier Transform of Periodic Time Domain Signals

$$\Lambda(t) = \begin{cases} 0 & ; |t| > 1 \\ 1-t & ; 0 \le t \le 1 \\ 1+t & ; -1 \le t \le 0 \end{cases}$$

a)



b) Determine & sketch the Fourier transform of $\Lambda(t)$

$$X(f) = \int_{-\infty}^{\infty} \Lambda(t) e^{-j2\pi ft}\,dt = \int_{-1}^{0} (1+t) e^{-j2\pi ft}\,dt + \int_{0}^{1} (1-t) e^{-j2\pi ft}\,dt$$

$$= \frac{1}{-j2\pi f}(1+t)e^{-j2\pi ft}\Big|_{-1}^{0} + \int_{-1}^{0} \frac{1}{+j2\pi f} e^{-j2\pi ft}\,dt + \frac{1-t}{-j2\pi f} e^{-j2\pi ft}\Big|_{0}^{1} + \int_{0}^{1} \frac{e^{-j2\pi ft}}{j2\pi f}\,dt$$

$$= \frac{1}{-j2\pi f} - 0 + \left[\frac{1}{4\pi^2 f^2} e^{-j2\pi ft}\right]_{-1}^{0} + 0 + \frac{1}{j2\pi f} + \left[\frac{1}{4\pi^2 f^2} e^{-j2\pi ft}\right]_{0}^{1}$$

$$= \frac{1}{4\pi^2 f^2} - \frac{e^{+j2\pi ft}}{4\pi^2 f^2} + \frac{e^{-j2\pi ft}}{4\pi^2 f^2} - \frac{1}{4\pi^2 f^2}$$

$$= \left(e^{-j2\pi ft} - e^{+j2\pi ft}\right) \frac{1}{4\pi^2 f^2}$$

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Fourier Transform of Periodic Time Domain Signals

b)

$$X(f) = \int_{-\infty}^{\infty} \Lambda(t) e^{-j2\pi ft}\,dt = \int_{-1}^{0} (1+t) e^{-j2\pi ft}\,dt + \int_{0}^{1} (1-t) e^{-j2\pi ft}\,dt$$

$$= \left[\frac{1+t}{-j2\pi f} e^{-j2\pi ft}\right]_{-1}^{0} - \int_{-1}^{0} \frac{e^{-j2\pi ft}}{-j2\pi f}\,dt + \left[\frac{1-t}{-j2\pi ft} e^{-j2\pi ft}\right]_{0}^{1}$$

$$- \int_{0}^{1} \frac{e^{-j2\pi ft}}{j2\pi f}\,dt$$

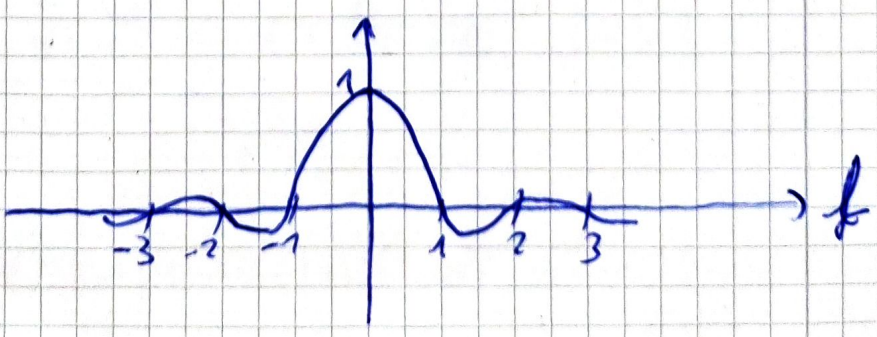
$$= \frac{1}{-j2\pi f} + \left[\frac{e^{-j2\pi ft}}{4\pi^2 f^2}\right]_{-1}^{0} + \frac{1}{+j2\pi f} - \left[\frac{e^{-j2\pi ft}}{4\pi^2 f^2}\right]_{0}^{1}$$

$$= \frac{1}{4\pi^2 f^2} - \frac{1}{4\pi^2 f^2} e^{j2\pi f} - \frac{1}{4\pi^2 f^2} e^{-j2\pi f} + \frac{1}{4\pi^2 f^2}$$

$$= \frac{1}{4\pi^2 f^2} \left(2 - \frac{1}{2}\left(e^{j2\pi f} + e^{-j2\pi f}\right)\right)$$

$$= \frac{1}{2\pi^2 f^2} \left(1 - \cos(2\pi f)\right)$$

$$= \frac{2\sin^2(\pi f)}{2\pi^2 f^2} = \underline{\underline{si^2(\pi f)}}$$



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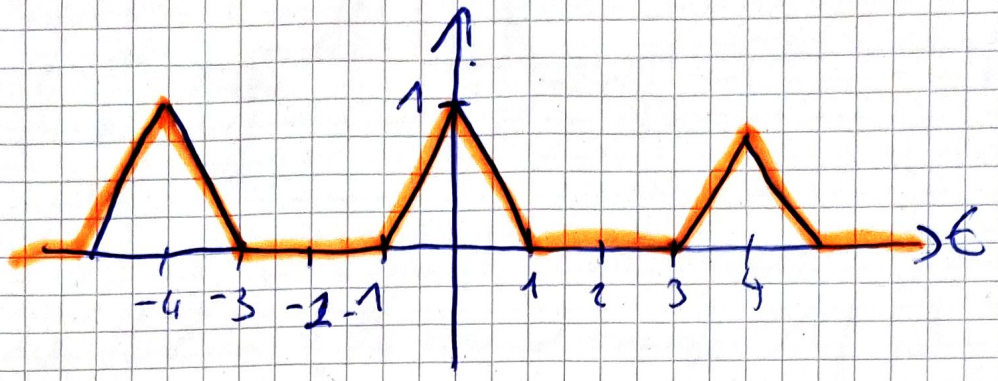
§ Fourier Transform of Periodic Time Signals

We know consider a periodic signal $x_T(t)$ which results from the periodic repetition of the signal with period $T = 4$

c)



d)

$$x_T(t) = \Lambda(t) * \sum_{k=-\infty}^{\infty} \delta(t - 4k)$$

e) $\Lambda(t) \circ\!\!-\!\!\bullet\; si^2(\pi f)$

$$\sum_{k=-\infty}^{\infty} \delta(t - 4k) \circ\!\!-\!\!\bullet\; \frac{1}{4} \sum_{-\infty}^{\infty} \delta\left(f - \frac{k}{4}\right)$$

$$X_T(f) = \frac{si^2(\pi f)}{4} \sum_{-\infty}^{\infty} \delta\left(f - \frac{k}{4}\right)$$

$$= \frac{1}{4} \sum_{-\infty}^{\infty} si^2\left(\pi \frac{k}{4}\right) \delta\left(f - \frac{k}{4}\right)$$

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$$x(t)\,\delta(t - t_0) = x(t_0)\,\delta(t - t_0)$$

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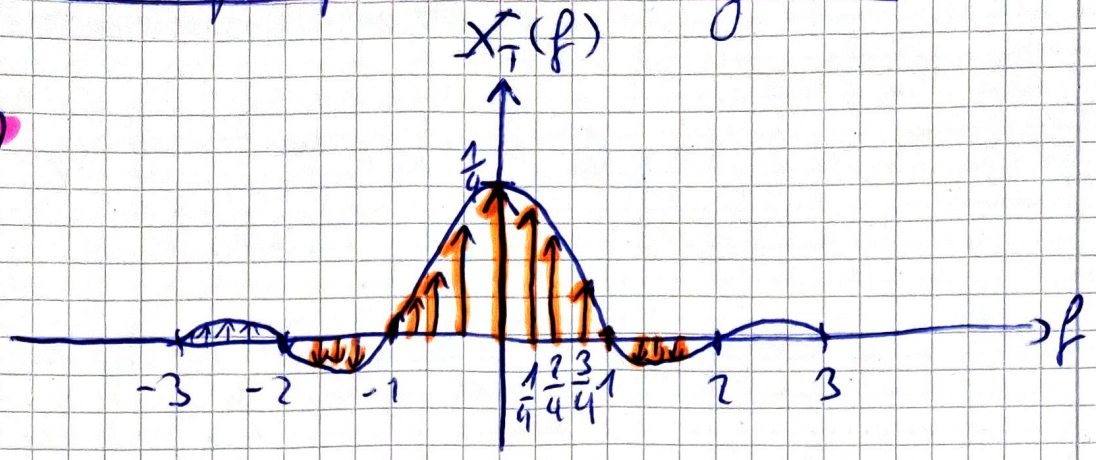
Fourier Transfan of Periodic Time D Signals

e)



f) identify the relation of the Fourier transfan $X(f)$ and the Fourier coefficients X_n of a Fourier series of the periodical signal $x_T(t)$

$$X_T(f) = \frac{1}{4} \sum_{-\infty}^{\infty} si^2\left(\pi \frac{k}{4}\right) \delta\left(f - \frac{k}{4}\right)$$

$$X_n = \frac{1}{4} si^2\left(\pi \frac{n}{4}\right)$$

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Fourier Series & Fourier Transform

a) $X(t) = \infty \cdot x_p(t) * \text{Ш}_T(t)$

b) $X(f) = X_p(f) \cdot \frac{1}{|T|} \sum_{n=-\infty}^{\infty} \delta(f - \frac{n}{T})$

$$= \frac{1}{|T|} \sum_{n=-\infty}^{\infty} X_p(\frac{n}{T}) \delta(f - \frac{n}{T})$$

c) $x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} dt$

$$= \frac{1}{|T|} \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} X_p(\frac{n}{T}) \delta(f - \frac{n}{T}) e^{j2\pi f t} dt$$

$$= \sum_{n=-\infty}^{\infty} \frac{1}{|T|} X_p(\frac{n}{T}) e^{j2\pi n/T t}$$

e) Def. d. Fourier series: $x(t) = \sum_{n=-\infty}^{\infty} x_n e^{j2\pi f_0 t}$

$f_0 = \frac{1}{T}$

$$x_n = \frac{1}{|T|} X_p(\frac{n}{T})$$

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